

Host Discovery and Enumeration

CompTIA PenTest+ (PT0-003)

Domain: 4.0: Attacks and Exploits

Objective: 4.2: Given a scenario, perform network attacks using the appropriate tools.

Video Overview

This video introduces essential active reconnaissance techniques crucial for penetration testing. You will learn how to identify live systems and gather information about their services and potential vulnerabilities. The concepts presented here are directly applicable to the CompTIA PenTest+ exam and will be valuable in your future cybersecurity career as you conduct security assessments and identify hidden threats in various network environments.

Key Topics Covered

- **Active Reconnaissance Overview:** Identifying live systems and gathering target information.
- **Host Discovery Techniques:** Using ping sweeps and Nmap for identifying active hosts.
- **Nmap Commands and Options:** Practical application of `-sn` and `-n` for efficient scanning.
- **Alternative Host Discovery Tools:** Zenmap, SolarWinds, and Angry IP Scanner.
- **Custom Scripting for Host Discovery:** Automating ping sweeps with Bash or Python.
- **Service Enumeration:** Identifying open ports, running services, and version information with Nmap (`-sV`).
- **Banner Grabbing:** Extracting service details using Telnet or Netcat.
- **Packet Crafting with Scapy:** Advanced manipulation of network packets for specialized tests.
- **War Driving:** Physical reconnaissance to discover unauthorized wireless access points ("shadow IT").
- **Packet Sniffing with Wireshark:** Capturing and analyzing network traffic for host discovery and sensitive information.

Student Notes Section